20

Tensors and Kronecker Products

Historical Note

The product in this chapter is named after Kronecker, who lectured on it in Berlin, but the determinant formula that first made it interesting was published in 1858 by Johann Georg Zehfuss, and the misattribution appears to have travelled through Kronecker's students rather than through anything he wrote. The word *tensor* has a separate and equally tangled history: Hamilton used it in 1846 for the magnitude of a quaternion, Voigt gave it its modern sense in elasticity around 1898, and Ricci and Levi-Civita built the calculus that Einstein later needed. What a machine learning library calls a tensor is none of those things exactly. It is a multi-way array of numbers, which is what Frank Hitchcock was studying in 1927 when he asked for the smallest number of rank-one terms needed to write one down. That question turns out to be far harder than the matrix case, and a good part of this chapter is about the size of the difference.

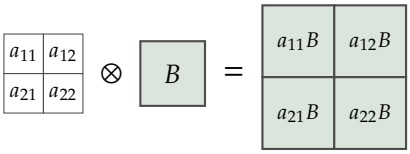
Chapter 19 described a large model entirely in terms of matrices, which was honest about the arithmetic and misleading about the shapes. Real data arrives with several indices at once: a batch of sequences of embeddings has three, a batch of images has four. This chapter supplies the algebra of several indices. The good news comes first, because the Kronecker product turns every question about it back into a question about ordinary matrices; the bad news is the last section, where genuinely multi-way notions such as rank stop behaving.

20.1 The Kronecker product

Definition 20.1 (Kronecker product)**.** *For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$, the* Kronecker

product $A \otimes B$ *is the* $mp \times nq$ *matrix built from* $m \times n$ *blocks, the* (i, j) *block being* $a_{ij}B$*:*

$$A \otimes B \;=\; \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$



every block is a scalar multiple of the whole of B, so the pattern of A is imprinted on a copy of B

Figure 20.1. Definition 20.1. The Kronecker product is not an entrywise operation and not a matrix product; it is the matrix of the map that applies A to one index and B to another at the same time. Its size is the product of the sizes, which is exactly why Section 20.3 insists that it never be written down.

Almost everything about the Kronecker product follows from one identity.

Theorem 20.2 (The mixed product rule)**.** *Whenever the ordinary products* AC *and* BD *are defined,*

$$(A \otimes B)(C \otimes D) \;=\; (AC) \otimes (BD).$$

Proof. Both sides are block matrices of the same shape, so it is enough to compare blocks. Multiplying block matrices as in Chapter 4, the (i, j) block of the left side is

$$\sum_k (a_{ik}B)(c_{kj}D) \;=\; \Big(\sum_k a_{ik}c_{kj}\Big)BD \;=\; (AC)_{ij}\,BD,$$

where the scalars a_{ik} and c_{kj} may be moved past the matrices freely. That is the (i, j) block of the right side. □

Corollary 20.3 (Consequences)**.** *Let* $A \in M_m(\mathbb{R})$ *and* $B \in M_p(\mathbb{R})$ *be square.*

(a) $(A \otimes B)^t = A^t \otimes B^t$*, and the same for the adjoint.*

(b) If A *and* B *are invertible then so is* $A \otimes B$*, with* $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$*.*

(c) If A *and* B *are orthogonal then so is* $A \otimes B$*.*

(d) $\operatorname{trace}(A \otimes B) = \operatorname{trace}(A)\operatorname{trace}(B)$.

(e) $\operatorname{rank}(A \otimes B) = \operatorname{rank}(A)\operatorname{rank}(B)$ *for rectangular* A, B *as well, and the singular values of* $A \otimes B$ *are the products* $\sigma_i(A)\sigma_j(B)$.

Proof. (a) is immediate from the definition, since transposing exchanges the block $a_{ij}B$ with $a_{ji}B^t$. For (b), the mixed product rule gives $(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = \mathbf{1}_m \otimes \mathbf{1}_p = \mathbf{1}_{mp}$, and (c) follows by combining (a) and (b). For (d), the diagonal blocks are $a_{ii}B$, so the trace is $\sum_i a_{ii}\operatorname{trace}(B)$.

For (e), take singular value decompositions $A = \sum_i \sigma_i u_i v_i^t$ and $B = \sum_j \tau_j w_j z_j^t$ from Proposition 16.6. The mixed product rule applied to vectors regarded as one-column matrices gives $(u \otimes w)(v \otimes z)^t = (uv^t) \otimes (wz^t)$, so

$$A \otimes B \;=\; \sum_{i,j} \sigma_i \tau_j \,(u_i \otimes w_j)(v_i \otimes z_j)^t .$$

The vectors $u_i \otimes w_j$ are orthonormal, because $\langle u_i \otimes w_j, u_k \otimes w_l \rangle = \langle u_i, u_k \rangle \langle w_j, w_l \rangle$ by the same rule, and likewise for the $v_i \otimes z_j$. So this is a singular value decomposition of $A \otimes B$, and the count of nonzero singular values is the product of the two counts. □

Theorem 20.4 (The spectrum of a Kronecker product)**.** *Let* $A \in M_m(\mathbb{C})$ *have eigenvalues* $\lambda_1, \dots, \lambda_m$ *and* $B \in M_p(\mathbb{C})$ *have eigenvalues* $\mu_1, \dots, \mu_p$*, both listed with algebraic multiplicity. Then the* mp *eigenvalues of* $A \otimes B$ *are the products* $\lambda_i \mu_j$*, again with multiplicity. If* $Ax = \lambda x$ *and* $By = \mu y$ *then* $x \otimes y$ *is an eigenvector for* $\lambda\mu$.

Proof. The statement about eigenvectors is one line of the mixed product rule:

$$(A \otimes B)(x \otimes y) \;=\; (Ax) \otimes (By) \;=\; (\lambda x) \otimes (\mu y) \;=\; \lambda\mu\,(x \otimes y).$$

For the full list with multiplicity, triangularise. By Theorem 14.5 write $A = UTU^*$ and $B = VSV^*$ with U, V unitary and T, S upper triangular carrying the eigenvalues on their diagonals. Then by the mixed product rule twice,

$$A \otimes B \;=\; (U \otimes V)\,(T \otimes S)\,(U \otimes V)^*,$$

and $U \otimes V$ is unitary by Corollary 20.3(c). Now $T \otimes S$ is upper triangular: its blocks below the diagonal vanish because the corresponding entries of T do, and each diagonal block $t_{ii}S$ is itself upper triangular. Its diagonal entries are the mp products $t_{ii}s_{jj}$. Since similar matrices have the same characteristic polynomial, by Proposition 9.6, those products are the eigenvalues of $A \otimes B$ with multiplicity. □

Corollary 20.5 (Determinant). *For $A \in M_m(\mathbb{R})$ and $B \in M_p(\mathbb{R})$, $\det(A \otimes B) = (\det A)^p\,(\det B)^m$.*

Proof. The determinant is the product of the eigenvalues, by Theorem 9.12. Each λ_i occurs in p of the products $\lambda_i\mu_j$ and each μ_j in m of them, so

$$\prod_{i,j} \lambda_i \mu_j \;=\; \Big(\prod_i \lambda_i\Big)^{p} \Big(\prod_j \mu_j\Big)^{m}. \qquad \square$$

Caution

Kronecker products do not commute: $A \otimes B$ and $B \otimes A$ are different matrices, as Figure 20.1 makes obvious. Theorem 20.4 says they have the same eigenvalues, and in fact more is true: they are permutation similar, so one is obtained from the other by reordering the indices. The exponents in Corollary 20.5 are also easy to get the wrong way round. The power on $\det A$ is the size of B, because each eigenvalue of A is repeated once for every eigenvalue of B.

20.2 Vectorisation, and the identity that does all the work

Figure 20.2 records what the last section bought. The spectrum of a Kronecker product is the set of all pairwise products of the two spectra, and of a Kronecker sum the set of all pairwise sums, so a grid of four entries is the complete answer for a pair of 2×2 matrices.



Figure 20.2. The spectrum of a Kronecker product is the set of all pairwise products of the two spectra, and of a Kronecker sum $A \otimes \mathbf{1} + \mathbf{1} \otimes B$ the set of all pairwise sums. With A having eigenvalues $1, 2$ and B having $3, 5$, the four entries of each grid are the complete answer, and the eigenvectors are the corresponding Kronecker products $u \otimes v$ in both cases. That is why separable problems separate: a discretised Laplacian on a rectangle is a Kronecker sum of one-dimensional Laplacians, so its eigenvalues are sums of known numbers and its eigenvectors are products of known vectors, and a two-dimensional problem is solved without ever forming a two-dimensional matrix.

Definition 20.6 (Vectorisation). *For $X \in \mathbb{R}^{m\times n}$ with columns $x_1, \dots, x_n$, write $\operatorname{vec}(X) = (x_1; x_2; \dots; x_n) \in \mathbb{R}^{mn}$ for the vector obtained by stacking the columns.*

Vectorisation is a linear isomorphism $\mathbb{R}^{m\times n} \to \mathbb{R}^{mn}$, and it is the bridge between the two-index world and the one-index world.

Theorem 20.7 (The vec identity). *For $A \in \mathbb{R}^{p\times m}$, $X \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times q}$,*

$$\operatorname{vec}(AXB) \;=\; \big(B^t \otimes A\big)\operatorname{vec}(X).$$

Proof. Two small facts first. If $u \in \mathbb{R}^p$ and $v \in \mathbb{R}^q$ then uv^t has jth column $v_j u$, so $\operatorname{vec}(uv^t) = v \otimes u$. And writing e_j for the standard basis of $\mathbb{R}^n$, the vector $e_j \otimes x_j$ has x_j in its jth block and zeros elsewhere, so $\sum_j e_j \otimes x_j = \operatorname{vec}(X)$.

Now expand $X = \sum_j x_j e_j^t$, so that

$$AXB \;=\; \sum_j (Ax_j)\,(e_j^t B) \;=\; \sum_j (Ax_j)\,(B^t e_j)^t.$$

Applying vec term by term with the first fact, and then the mixed product rule,

$$\operatorname{vec}(AXB) \;=\; \sum_j (B^t e_j) \otimes (Ax_j) \;=\; \sum_j (B^t \otimes A)(e_j \otimes x_j) \;=\; (B^t \otimes A)\operatorname{vec}(X). \qquad \square$$

Key Idea

Read the identity from right to left and it says something useful about modelling. A linear map on matrix-shaped data is in general an arbitrary $mn \times mn$ matrix, with m^2n^2 free parameters. The maps of the special form $X \mapsto AXB$, which act on the two indices separately, are exactly the Kronecker operators, and they have only $m^2 + n^2$ parameters. That is the entire content of the phrase *separable*, whether it is applied to a convolution, to a transform or to a covariance model. Chapter 19 treated a batch of inputs as the columns of a matrix; the statement that a layer acts on each column independently is the statement that its matrix is $\mathbf{1}_n \otimes A$.

Proposition 20.8 (The Kronecker sum). *For $A \in M_m(\mathbb{C})$ and $B \in M_n(\mathbb{C})$ with eigenvalues λ_i and μ_j, the mn eigenvalues of $\mathbf{1}_n \otimes A + B^t \otimes \mathbf{1}_m$ are the sums $\lambda_i + \mu_j$.*

Proof. Triangularise both factors as in Theorem 20.4, writing $A = UTU^{-1}$ and $B^t = VSV^{-1}$; note B^t has the same eigenvalues as B by Proposition 9.18. Conjugating by

$V \otimes U$ and using the mixed product rule on each summand,

$$(V \otimes U)^{-1}\big(\mathbf{1}_n \otimes A + B^t \otimes \mathbf{1}_m\big)(V \otimes U) \;=\; \mathbf{1}_n \otimes T + S \otimes \mathbf{1}_m,$$

which is upper triangular with diagonal entries $t_{ii} + s_{jj}$. □

Example 20.9 (Sylvester's equation). *The matrix equation $AX + XB = C$, with A of size m, B of size n and X unknown, looks unlike anything in Chapter 3 until it is vectorised. By Theorem 20.7 with the two special cases $\operatorname{vec}(AX) = (\mathbf{1}_n \otimes A)\operatorname{vec}(X)$ and $\operatorname{vec}(XB) = (B^t \otimes \mathbf{1}_m)\operatorname{vec}(X)$, it becomes the ordinary linear system*

$$\big(\mathbf{1}_n \otimes A + B^t \otimes \mathbf{1}_m\big)\operatorname{vec}(X) \;=\; \operatorname{vec}(C)$$

in mn unknowns. By Proposition 20.8 the coefficient matrix is invertible exactly when no $\lambda_i + \mu_j$ vanishes, so:

$AX + XB = C$ has a unique solution for every C if and only if A and $-B$ have no eigenvalue in common.

That is Sylvester's theorem of 1884. The case $B = A^t$ is the Lyapunov equation $AX + XA^t = C$ of stability theory, which is uniquely solvable exactly when no two eigenvalues of A sum to zero.

20.3 Never form the Kronecker product

The saving in Figure 20.3 is worth stating as an identity rather than a slogan. Applying $B^t \otimes A$ to $\operatorname{vec}(X)$ means computing AXB, and AXB is computed as $(AX)B$: two ordinary matrix products, costing m^2n and mn^2 multiplications by the count of Proposition 4.7, so $mn(m+n)$ in total against m^2n^2 for the literal matrix-vector product. The ratio is $mn/(m+n)$, which at $m = n$ is $n/2$.

Insight

This is the lesson of Chapter 17 in its purest form: structure is the only thing that ever makes a computation cheaper, and a structure is worth having exactly when it can be exploited without ever being written down. Every high-performance library implements Kronecker operators as a pair of matrix multiplications, and every framework that offers a `batched matmul` is offering $\mathbf{1} \otimes A$ under another name. The reason such an operation is a single library call rather than a loop is that Chapter 4 already showed blocking to be arithmetically free, so the batch can be laid out to keep the hardware busy.

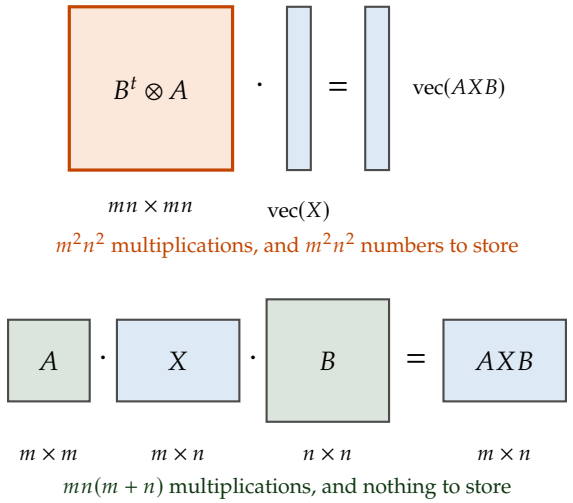


Figure 20.3. Theorem 20.7 read as advice. The two rows compute the same numbers. At $m = n = 1000$ the upper route needs 10^{12} multiplications and eight terabytes of storage for a matrix every entry of which is a product of two numbers already in memory; the lower route needs 2×10^9 multiplications and no storage at all. The Kronecker product is a tool for proving things, not for computing them.

Example 20.10 (Kronecker-factored curvature). *Proposition 19.5 showed that for a layer $y = Wx$ the gradient is the outer product $g\,x^t$, where $g = \nabla_y \ell$. Vectorising it, $\mathrm{vec}(gx^t) = x \otimes g$. Second-order optimisation methods want the matrix*

$$F \;=\; \mathsf{E}\big[(x \otimes g)(x \otimes g)^t\big] \;=\; \mathsf{E}\big[xx^t \otimes gg^t\big],$$

the second equality being the mixed product rule again. If W is $d \times d$ then F is $d^2 \times d^2$, so at $d = 1000$ it has 10^{12} entries and inverting it would cost about 10^{18} operations. The Kronecker-factored approximation replaces the expectation of the product by the product of the expectations,

$$F \;\approx\; \mathsf{E}[xx^t] \otimes \mathsf{E}[gg^t] \;=\; A \otimes G,$$

after which Corollary 20.3(b) gives $F^{-1} \approx A^{-1} \otimes G^{-1}$: two $d \times d$ inversions, about 2×10^9 operations, and the result is applied to a gradient by Theorem 20.7 without ever being assembled.

Caution

The step $\mathsf{E}[xx^t \otimes gg^t] \approx \mathsf{E}[xx^t] \otimes \mathsf{E}[gg^t]$ is not an identity and not an approximation with a proved error bound. It is exact when the input to a layer and the gradient arriving at its output are independent, which they are not. Everything else in Example 20.10 is a theorem; that one line is a modelling assumption, and the method works to the extent that it happens to be nearly true. It is worth separating the two, because the algebra is often quoted as though it justified the whole method.

20.4 Tensors as multi-way arrays

Definition 20.11 (Tensor, fibres and unfoldings)**.** *An* order-d tensor *is an array* $T \in \mathbb{R}^{n_1 \times \cdots \times n_d}$*, written* $T_{i_1 \ldots i_d}$*. Order* 1 *is a vector and order* 2 *a matrix. Fixing all indices but the* k*th gives a* mode-k fibre*. The* mode-k unfolding $T_{(k)}$ *is the* $n_k \times \prod_{i \neq k} n_i$ *matrix whose columns are the mode-*k *fibres.*

Caution

In differential geometry a tensor is a multilinear map together with a rule for how its coordinates change under a change of basis, and the array is only its representation in a chosen basis, exactly as a matrix is the representation of a linear map in Chapter 7. In this chapter, and in every machine learning library, *tensor* means the array itself. The two usages are related in the same way as *matrix* and *linear map* and are confused about as often.

Figure 20.4 shows the three unfoldings of one array. They hold the same $n_1 n_2 n_3$ numbers, so the three rectangles have equal area and different shapes; what changes is which pairs of entries the matrix theory of the previous nineteen chapters can see.

Write $a \circ b \circ c$ for the order-3 array with entries $a_i b_j c_k$, and call such an array, when all three vectors are nonzero, a *rank-one* tensor. For two factors the notation agrees with the outer product of Chapter 16: $a \circ b$ is the matrix ab^t.

Caution

The symbols $\otimes$ and $\circ$ are doing different jobs and both are standard. For vectors $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$, the Kronecker product $a \otimes b$ is a *vector* of length mn, while $a \circ b$ is an $m \times n$ *matrix*. They hold the same numbers, related by Theorem 20.7's

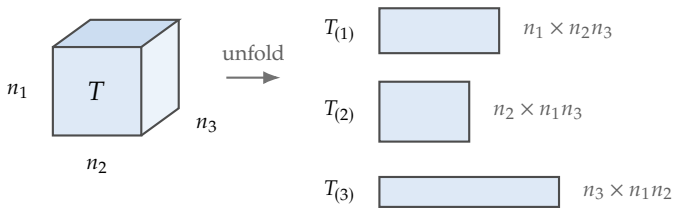


Figure 20.4. Definition 20.11. Unfolding rearranges the same $n_1 n_2 n_3$ numbers into a matrix, so the three rectangles have equal area and different shapes. Nothing is lost and nothing is gained: what changes is which pairs of entries the matrix theory of the previous nineteen chapters is able to compare. The whole of Section 20.5 is about what that theory can and cannot see once the array has three indices.

first step: $\operatorname{vec}(a \circ b) = b \otimes a$. Note the reversal of order, which is the commonest slip in this subject.

20.5 Tensor rank, and four ways it misbehaves

Definition 20.12 (Tensor rank)**.** *The* rank *of a tensor T is the smallest R for which*

$$T = \sum_{r=1}^{R} a_r \circ b_r \circ c_r,$$

an expression called a CP decomposition*. The* multilinear rank *is the triple* $\big(\operatorname{rank} T_{(1)}, \operatorname{rank} T_{(2)}, \operatorname{rank} T_{(3)}\big)$.

For matrices, Definition 20.12 reproduces the ordinary rank, and the two notions coincide. For order 3 they are different, and the first is badly behaved in four separate ways.

Remark 20.13 (Four ways tensor rank misbehaves)**.** *Let T be an order-3 tensor.*

(a) Rank depends on the field. *The $2 \times 2 \times 2$ tensor whose two frontal slices are $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has rank 3 over $\mathbb{R}$ and rank 2 over $\mathbb{C}$. Nothing like this happens for matrices, whose rank is the number of nonzero pivots and so is unchanged by enlarging the field.*

(b) The set of tensors of rank at most R is not closed. *Take a, a' independent, and*

likewise b, b' and c, c', and set

$$T_\varepsilon \;=\; \tfrac{1}{\varepsilon}\Big[(a+\varepsilon a')\circ(b+\varepsilon b')\circ(c+\varepsilon c') \;-\; a\circ b\circ c\Big].$$

Each T_ε is a difference of two rank-one tensors, so has rank at most 2; expanding and letting $\varepsilon \to 0$ gives $a'\circ b\circ c + a\circ b'\circ c + a\circ b\circ c'$, which has rank 3.

(c) The best low-rank approximation need not exist. *By (b) the infimum of $\|T-S\|$ over tensors S of rank at most 2 can fail to be attained, so the optimisation problem that Eckart and Young solved for matrices in Theorem 16.12 may have no solution at all.*

(d) Computing the rank is hard. *Deciding the rank of a given tensor is NP-hard, and so are most other natural questions about tensors.*

Key Idea

It is worth being precise about which chapter is being lost. The singular value decomposition gave three things at once: a factorisation that always exists, a rank that is easy to compute, and a best low-rank approximation that is optimal and essentially unique. For order 3 the first survives in a weakened form, the second fails, and the third can fail so completely that the problem is ill-posed. This is not a gap in the literature waiting to be filled. It is a genuine discontinuity between two and three indices, and it is the reason that practical work with tensors is done by unfolding them into matrices.

Theorem 20.14 (What survives: the truncated HOSVD)**.** *Let $T \in \mathbb{R}^{n_1\times\cdots\times n_d}$ and let U_k hold the leading r_k left singular vectors of the unfolding $T_{(k)}$, from Theorem 16.3. Let $\widehat{T}$ be the projection of T onto the span of these in every mode. Then $\widehat{T}$ has multilinear rank at most $(r_1,\dots,r_d)$ and*

$$\left\|T-\widehat{T}\right\|_F \;\le\; \sqrt{d}\;\min\left\{\|T-S\|_F : S \text{ of multilinear rank at most } (r_1,\dots,r_d)\right\}.$$

Proof. For each k let P_k be the map on tensors that replaces every mode-k fibre by its orthogonal projection onto span of the columns of U_k; in unfolded form, $(P_kT)_{(k)} = U_kU_k^tT_{(k)}$. Projections in different modes act on different indices, so $P_kP_l = P_lP_k$, and $\widehat{T} = P_1P_2\cdots P_dT$.

Each mode separately. By Proposition 13.6(a), $U_kU_k^t$ is the orthogonal projection onto the span of the leading r_k left singular vectors of $T_{(k)}$, so $U_kU_k^tT_{(k)}$ is the rank-r_k truncated

singular value decomposition of $T_{(k)}$ and

$$\|T - P_k T\|_F^2 = \sum_{i > r_k} \sigma_i\big(T_{(k)}\big)^2$$

by Theorem 16.12, the sum being empty, and both sides zero, when r_k is already at least rank $T_{(k)}$. Now let S have multilinear rank at most $(r_1, \dots, r_d)$, so rank $S_{(k)} \le r_k$. Unfolding changes no entry, so $\|T - S\|_F = \big\|T_{(k)} - S_{(k)}\big\|_F$, and the Frobenius half of Theorem 16.12 bounds that below by the same tail of singular values. Hence $\|T - P_k T\|_F \le \|T - S\|_F$ for every k.

Combining the modes. Telescoping,

$$T - \widehat{T} \;=\; \sum_{k=1}^{d} P_1 \cdots P_{k-1}\big(I - P_k\big)T,$$

with the empty product read as the identity. These d summands are pairwise orthogonal: for $k < l$ the kth lies in the range of $I - P_k$ while the lth, carrying the factor P_k among its leading projections, lies in the range of P_k. So

$$\Big\|T - \widehat{T}\Big\|_F^2 = \sum_{k=1}^{d} \|P_1 \cdots P_{k-1}(I - P_k)T\|_F^2 \;\le\; \sum_{k=1}^{d} \|(I - P_k)T\|_F^2 \;\le\; d\;\|T - S\|_F^2\,,$$

the middle step because an orthogonal projection never increases the Frobenius norm. Taking the square root and then the minimum over S gives the claim. Finally $\widehat{T}$ has multilinear rank at most $(r_1, \dots, r_d)$ because every mode-k fibre of $\widehat{T}$ lies in the span of the r_k columns of U_k, so rank $\widehat{T}_{(k)} \le r_k$. □

So the answer is not optimal, but it is within a fixed factor of optimal, it is computed by d ordinary singular value decompositions, and it always exists. In practice that is worth a great deal more than an optimum that may not.

Example 20.15 (A separable operator)**.** *A blur that acts on the rows and the columns of an image independently is the map $X \mapsto AXB$. For a 1024×1024 image the general linear map on that space has $1024^4 \approx 1.1 \times 10^{12}$ parameters, while the separable one has $2 \times 1024^2 \approx 2.1 \times 10^6$: a factor of half a million. If in addition A and B are banded, as they are for a local blur, the count drops again. Every claim about what such an operator does, its spectrum, its invertibility, its condition number, is answered by Corollary 20.3 and Theorem 20.4 in terms of A and B separately. For instance its condition number in the spectral norm is the product $\kappa(A)\kappa(B)$, by Corollary 20.3(e).*

Open Problem 20.16 (The maximum rank of a small tensor)**.** *Every $n \times n$ matrix has rank at most n, and that bound is attained and easy. For tensors the analogous question is open in almost every case.* What is the maximum rank of an $n \times n \times n$ tensor over $\mathbb{C}$?
What is known. *The generic rank, the rank of almost every tensor, is known to be $\lceil n^3/(3n-2) \rceil$ for most n, by a dimension count made rigorous by Lickteig. The* maximum *rank is larger and is known only for $n \le 3$: for $2\times2\times2$ it is 3 and for $3\times3\times3$ it is 5. For $4\times4\times4$ it is known only to lie between 7 and 10. Even the question of whether the maximum rank is asymptotically $n^2/3$ or larger is open.*
Why matrices give no guidance. *Every fact this chapter's readers rely on for matrices fails here. Rank can increase under limits, so the set of tensors of rank at most r is not closed, which is Chapter 101's subject. Rank over $\mathbb{R}$ can exceed rank over $\mathbb{C}$. And rank is not additive over direct sums: Strassen conjectured it was for three decades, and Shitov produced counterexamples in 2019 [565].* Every structural theorem that makes matrix rank easy is false one dimension up.
Where a reader might start. *Show by hand that the $2 \times 2 \times 2$ tensor of matrix multiplication, which is $\langle 2,2,2 \rangle$, has rank at most 7, which is Strassen's algorithm, and rank at least 7, which is a short argument. Then find a $2 \times 2 \times 2$ tensor of rank 3 and prove it cannot be 2; the standard example is the tensor with a single one in three positions forming an "L". Once rank 3 is real, the phrase "maximum rank exceeds the format" stops being surprising and becomes a question.*

Exercises

Exercise 20.17. *For $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ write out $A \otimes B$ and $B \otimes A$ in full. Verify that they are different, that they have the same trace and determinant, and that both determinants agree with Corollary 20.5.*

Exercise 20.18. *Use Theorem 20.2 to prove that $(A \otimes B)^k = A^k \otimes B^k$ for every $k \ge 1$, and that $A \otimes B$ is symmetric whenever A and B are. Is the converse of the second statement true?*

Exercise 20.19. *Let A have eigenvalues $2, -1$ and B have eigenvalues $3, 3, 0$. Write down the six eigenvalues of $A \otimes B$, the trace and the determinant, and check the last two against Corollary 20.3(d) and Corollary 20.5. Now write down the eigenvalues of $\mathbf{1}_3 \otimes A + B^{t} \otimes \mathbf{1}_2$.*

Exercise 20.20. *Prove that* $\operatorname{vec}(uv^{t}) = v \otimes u$ *directly from Definition 20.6, and deduce Theorem 20.7 in the special case where* X *has rank one. Why does the general case follow?*

Exercise 20.21. *Solve* $AX + XB = C$ *for* $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ *and* $C = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ *by vectorising. Then find a* B *for which the equation is not solvable for every* C*, and say which eigenvalues collide.*

Exercise 20.22. *Take* $m = n = 512$*. Count the multiplications needed to compute* AXB *by the two routes of Figure 20.3, and compute how many gigabytes* $B^{t} \otimes A$ *would occupy in double precision. Repeat for* $m = n = 4096$ *and comment.*

Exercise 20.23. *Let* A *and* B *both be* 2×2*. Find explicitly a permutation matrix* P *with* $P(A \otimes B)P^{t} = B \otimes A$*, by writing down what* P *must do to the index pairs. Explain why such a* P *exists in general, and why this gives a second proof that the two products have the same eigenvalues.*

Exercise 20.24. *Let* T *be the* $2 \times 2 \times 2$ *tensor with* $T_{ijk} = i + j + k$*. Write down its three unfoldings and check that the multilinear rank is* $(2, 2, 2)$*. Then verify that*

$$T \;=\; a \circ w \circ w \;+\; w \circ a \circ w \;+\; w \circ w \circ a, \qquad a = (1, 2), \quad w = (1, 1),$$

so the rank is at most 3*. It is in fact exactly* 3*, so the rank exceeds every one of the multilinear ranks. Explain why nothing like that can happen for a matrix.*

Exercise 20.25. *Expand* T_{ε} *from Remark 20.13(b) in powers of* ε *and confirm that the limit as* $\varepsilon \to 0$ *is the stated tensor. Explain in one sentence why this makes the phrase "the best rank-two approximation" meaningless for some tensors, and contrast with what Theorem 16.12 guarantees for matrices.*

References and further reading

Van Loan's survey article on the ubiquitous Kronecker product [429] is the best short account of Section 20.3 and collects the identities of Corollary 20.3 with their applications; Golub and Van Loan [263] treat the same material alongside the algorithms, and are also the reference for the numerical solution of Sylvester and Lyapunov equations, which

is not done by vectorising but by first reducing A and B to Schur form. Henderson, Pukelsheim and Searle [317] settled the attribution question in the historical note. For tensors, the survey of Kolda and Bader [379] is the standard entry point and is where the unfolding notation of Definition 20.11 comes from; the four failures of Remark 20.13 are due respectively to Kruskal, to De Silva and Lim [568], again to De Silva and Lim, and to Håstad, with Hillar and Lim [326] showing how general the last one is. The quasi-optimality bound of Theorem 20.14 is due to De Lathauwer, De Moor and Vandewalle [404]. Banica [51], our baseline, does not treat tensors, and this chapter is an addition of our own.